

Plant Incident Report INC-2022-0820

P-101 non-drive-end bearing seizure | 2022-08-20 | Unit-2 Utilities Block

Incident number	INC-2022-0820	Date of event	2022-08-20
Equipment	P-101 — Boiler Feed Water Pump P-101	Failure mode	NDE bearing seizure
Classification	Category 2 — significant production loss, no injury	Reportable	No injury; not reportable under Section 88
Downtime	14 hours	Total cost	Rs 25,00,000 (Rs 25 lakh)
Investigation lead	Priya Sharma	RCA reference	RCA-2022-0825

1. Summary

At 04:52 on 2022-08-20 the non-drive-end bearing of P-101 seized while the machine was running at rated duty. The shaft locked, the motor tripped on overcurrent, and Unit-2 lost boiler feed water supply. The standby machine P-102 was started manually at 05:04 but tripped on high vibration; steam raising was lost for a total of 14 hours until P-101 was returned to service with a new bearing set.

2. Timeline

Date / time	Event
2022-08-14	NDE bearing temperature 68 degC at routine round. Recorded, no action.
2022-08-19 22:40	VT-1011 high vibration alarm (8.1 mm/s). Acknowledged on the night shift. No work order raised.
2022-08-19 23:55	VT-1011 high vibration alarm again (8.9 mm/s). Acknowledged. No corrective action.
2022-08-20 02:30	Lube oil header pressure 1.2 bar, below the 1.4 bar low-low limit. No alarm configured on PI-1016 at the time.
2022-08-20 04:52	NDE bearing seized. Shaft locked. Motor tripped on overcurrent.
2022-08-20 05:04	P-102 started manually; tripped on high vibration after 6 minutes.
2022-08-20 05:20	Unit-2 steam raising lost. Deaerator V-201 level held on condensate return.
2022-08-20 18:52	P-101 returned to service after bearing replacement. Total downtime 14 hours.

3. Precursors identified in the investigation

1. NDE bearing temperature rose from 68 degC to 91 degC over the six days preceding the seizure.
2. Overall vibration at the NDE bearing trended from 4.2 mm/s to 8.9 mm/s RMS in the same window.
3. Lube oil header pressure decayed from 2.1 bar to 1.2 bar, below the 1.4 bar low-low limit.
4. Two VT-1011 high-vibration alarms were acknowledged on the night shift without corrective action.
5. Bearing wear was recorded at 84% of the OEM replacement limit at the June 2022 inspection.

Each precursor was individually visible in plant records before the failure. None was individually alarming. The combination was never assembled by any single person or system, which is the central finding of this report.

4. Consequences and cost

Cost element	Amount (Rs)	Basis
Lost steam production	14,20,000	14 h at the Unit-2 contribution rate
Emergency bearing set and freight	4,80,000	Air freight from the OEM
Contract labour, 3 shifts	2,60,000	Rotating equipment crew
Shaft skim and re-metalling	2,15,000	External workshop
Lube oil flush and refill	1,25,000	ISO VG 46, 220 I
Total	25,00,000	Rs 25,00,000 (Rs 25 lakh)

Total cost Rs 25,00,000 (Rs 25 lakh) against 14 hours of downtime. No injury occurred. There was no loss of containment and no environmental consequence.

5. Immediate actions taken

- P-101 isolated and the NDE and DE bearing sets replaced per SOP-MECH-014 by Rajesh Kumar.
- LP-101A auxiliary lube oil pump found with heavy internal wear and overhauled on 2022-08-24; discharge restored to 2.1 bar.
- P-102 standby machine inspected; alignment corrected and returned to standby duty.
- Lube oil system flushed and recharged; oil sample sent for analysis.
- Root cause analysis RCA-2022-0825 commissioned; issued 2022-08-25.

6. Equipment affected

Tag	Description	Role in the event
P-101	Boiler Feed Water Pump P-101	Failed machine — NDE bearing seizure
LP-101A	Auxiliary Lube Oil Pump LP-101A	Root cause — lube oil pressure decay
P-102	Boiler Feed Water Pump P-102 (standby)	Standby — tripped on start, extended the outage
V-201	Deaerator Storage Vessel V-201	Suction source — level held on condensate return

Prepared by Priya Sharma. Reviewed by D. Krishnan. Witness statements from Rajesh Kumar and Anil Deshmukh are held in the investigation file.